Limits and Derivatives

Unit 3-1 Lecture Notes

MSDS 520: Math Fundamentals for Data Science
Meharry Medical College

These notes are the written companion to the Unit 3-1 slides and follow them section by section. They go further than the slides: every idea comes with extra worked examples, the traps students most often fall into are flagged in boxes titled **Watch out**, and several sections end with additional practice. Every exercise is followed by a worked solution; try each one before reading on.

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1 Calculus: the Study of Change

1.1 A Problem to Motivate Us

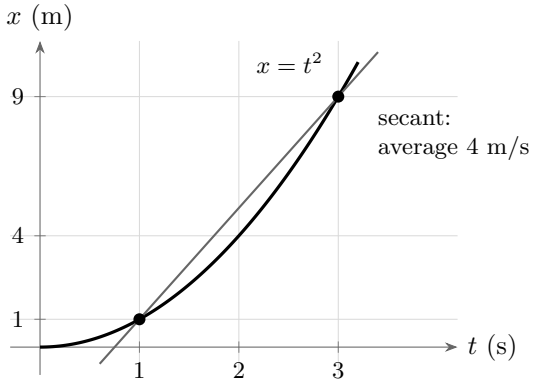
A car pulls away from a stop sign. Its position, in meters, t seconds after it starts is

$$x(t) = t^2.$$

A few readings:

time t (s)	0	1	2	3
position $x(t)$ (m)	0	1	4	9

From $t = 1$ to $t = 3$ the car moves from 1 m to 9 m: 8 meters in 2 seconds, an **average** speed of $8/2 = 4$ m/s. From $t = 1$ to $t = 2$ it covers 3 m in 1 s, an average of 3 m/s. The car is speeding up, so the average depends on which stretch of time we pick.



Now the real question. A speedometer does not report an average over two seconds; it reports the speed **at one instant**. What does it read at exactly $t = 1$?

Speed is distance divided by time, and that needs **two** instants. If we use the single instant $t = 1$ as both the start and the end, the car covers 0 meters in 0 seconds, and

$$\text{speed} = \frac{0}{0},$$

which is not a number. Yet the car obviously *has* a speed at $t = 1$; the needle points somewhere.

This lecture's question

How do we measure a rate of change at a **single instant**? The answer is the **derivative**, and building it is the job of these notes.

1.2 Where This Shows Up in Data Science

A **rate of change** answers "how fast is one quantity changing as another one changes?" They are everywhere:

- **Velocity**: how fast position changes with time, in meters per second.
- **Drug elimination**: how fast a drug's concentration in the blood falls, in mg/L per hour. Dosing schedules are built on this rate.
- **Epidemics**: new cases per day is a rate. Public health officials also watch whether that number is itself growing, which is a rate of a rate.
- **Temperature across a room**: degrees per meter. A rate can be with respect to **position**, not just time.

For a data scientist, the most important rate is this one. A model has adjustable numbers called **weights**, and it makes some amount of **error** on the data. Training the model means asking, over and over: *if I nudge this weight a little, how fast does the error change, and in which direction?*

That rate is a derivative. Every model you will train, from linear regression to a neural network, is trained by following derivatives downhill. Unit 3-2 does exactly that; these notes build the tool.

1.3 What We Will Build

1. The **average** rate of change: secant lines and the **difference quotient**.
2. Shrink the interval, and learn **just enough** about limits to get past 0/0.
3. The **derivative**: its definition, its notation, and six examples worked from the definition.
4. **Rules** (power, sum, product, quotient): shortcuts, so we never have to take the limit by hand again.
5. The **chain rule**, for functions inside functions. Every error function in Unit 3-2 needs it.
6. **Partial derivatives**: for a function of two variables (Unit 1-3), a slope in each direction.

Five notebooks go with these notes, and pointers to the relevant section of each appear throughout:

- `U3-2_Limits-1_ApproachingALimit.ipynb`
- `U3-3_Derivatives-1_SecantToTangent.ipynb`
- `U3-3_Derivatives-2_Rules.ipynb`
- `U3-6_Multivariable-1_Surface3D.ipynb`
- `U3-6_Multivariable-2_PartialDerivatives.ipynb`

2 Average Rate of Change

2.1 Rate of Change Is Rise over Run

Definition: average rate of change

The **average rate of change** of a function f from x_1 to x_2 is

$$\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{\text{rise}}{\text{run}}.$$

The symbol Δ (Greek capital "delta") means "change in": Δy is the change in output, Δx the change in input. The units are **output units per input unit**.

The sign carries meaning. A positive rate means the output went **up** as the input increased; a negative rate means it went **down**; a rate of 0 means it ended where it started.

Worked example: the car. From $t = 1$ to $t = 3$,

$$\frac{x(3) - x(1)}{3 - 1} = \frac{9 - 1}{2} = 4 \text{ m/s}.$$

The output is in meters and the input in seconds, so the rate is in meters per second.

Worked example: a drug clearing the blood. A patient's blood concentration of a drug, C, falls from 8 mg/L to 2 mg/L over 3 hours. Then

$$\frac{\Delta C}{\Delta t} = \frac{2-8}{3} = \frac{-6}{3} = -2 \text{ mg/L per hour.}$$

The rate is negative because the concentration is **falling**: on average it lost 2 mg/L every hour.

Worked example: a function with no story. For $f(x) = x^2$ from $x_1 = -2$ to $x_2 = 0$,

$$\frac{f(0) - f(-2)}{0 - (-2)} = \frac{0-4}{2} = -2.$$

On the left half of the parabola the graph comes **down** toward the bottom, so the rate is negative.

Worked example: weight loss. A patient weighs 92 kg at their January checkup and 86.5 kg five months later. The average rate of change is

$$\frac{86.5 - 92}{5} = \frac{-5.5}{5} = -1.1 \text{ kg per month.}$$

Watch out

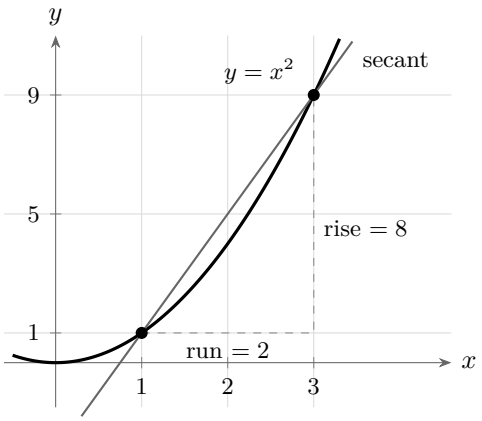
- **Subtract in the same order on top and bottom.** Whichever point you call "second" comes first in *both* the numerator and the denominator. Writing $\frac{f(x_1)-f(x_2)}{x_2-x_1}$ flips the sign of the answer.
- **An average hides the path.** The patient above might have lost 8 kg and regained 2.5; the average rate only sees the two endpoints.

2.2 Secant Lines

The average rate of change has a picture.

Definition: secant line

A **secant line** is a line that crosses a graph at **two** points. Its slope is the **average** rate of change between those two points.



On $y = x^2$, the secant through $(1, 1)$ and $(3, 9)$ rises 8 over a run of 2, so its slope is $8/2 = 4$: the car's average speed, drawn.

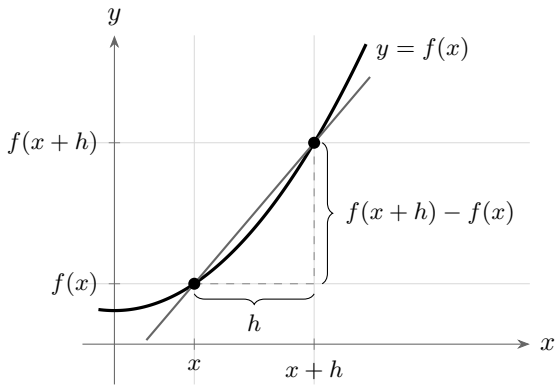
The catch. The curve is plainly steeper near $x = 3$ than near $x = 1$, but the secant reports one number for the whole stretch. Split the stretch in half and the difference shows:

$$\text{from 1 to 2: } \frac{4-1}{1} = 3, \qquad \text{from 2 to 3: } \frac{9-4}{1} = 5.$$

The two halves have slopes 3 and 5, and their average is the 4 we had before. The shorter the stretch, the more local the information. That observation drives the rest of these notes.

2.3 The Difference Quotient

We now rename the two points so that the formula is about **one** point and a **step** away from it. Start at x and take a step of size h to $x + h$. The two points on the graph are $(x, f(x))$ and $(x + h, f(x + h))$.



Definition: difference quotient

$$\frac{f(x+h) - f(x)}{h}$$

- h is the **run**: the size of the step.
- $f(x + h) - f(x)$ is the **rise**.
- The quotient is the slope of the secant from x to $x + h$.

This is the same rise-over-run formula as before, with $x_1 = x$ and $x_2 = x + h$, so that $x_2 - x_1 = h$. Nothing new has happened yet except the names. The point of the new names: h is now a single dial we can turn, and we are about to turn it toward zero.

Worked example: $f(x) = x^2$ at $x = 1$, for any step h. First find $f(1 + h)$ by expanding:

$$f(1+h) = (1+h)^2 = 1 + 2h + h^2.$$

Then

$$\frac{f(1+h) - f(1)}{h} = \frac{1 + 2h + h^2 - 1}{h} = \frac{2h + h^2}{h} = \frac{h\,(2+h)}{h} = 2 + h.$$

The cancellation is allowed because h is a step, so $h \neq 0$. One formula now gives every secant slope from $x = 1$ at once: $h = 2$ gives 4 (the secant from 1 to 3), $h = 1$ gives 3 (from 1 to 2), $h = 0.5$ gives 2.5.

Worked example: a line. For $f(x) = 5x - 1$,

$$\frac{5(x+h) - 1 - (5x - 1)}{h} = \frac{5x + 5h - 1 - 5x + 1}{h} = \frac{5h}{h} = 5.$$

Every secant of a line has the line's own slope, whatever the step.

Watch out: computing $f(x + h)$

$f(x + h)$ means "replace **every** x in the formula for f by $(x + h)$, in parentheses". It is **not** $f(x) + h$, and it is **not** $f(x) + f(h)$. For $f(x) = x^2$:

$$f(x+h) = (x+h)^2 = x^2 + 2xh + h^2, \qquad \text{not } x^2 + h \text{ and not } x^2 + h^2.$$

The middle term $2xh$ is the one most often lost, and it is the one that matters.

Exercise 2.1

1. For $f(x) = x^2$, compute the difference quotient at $x = 1$ with $h = 1$, then with $h = 0.5$. What is happening to the slope?
2. A patient's temperature is 101.3°F at 8 am and 99.5°F at 2 pm. Find the average rate of change, with units, and say what it means.
3. For the line $f(x) = 3x + 2$, simplify $\dfrac{f(x+h) - f(x)}{h}$ for **any** h. What do you notice?

Solution to Exercise 2.1

1. With $h = 1$ the second point is $x + h = 2$:

   $$\frac{f(2) - f(1)}{1} = \frac{4 - 1}{1} = 3.$$

   With $h = 0.5$ the second point is 1.5, and $f(1.5) = 2.25$:

   $$\frac{f(1.5) - f(1)}{0.5} = \frac{2.25 - 1}{0.5} = \frac{1.25}{0.5} = 2.5.$$

   As h shrinks the secant gets **less steep**: it is settling toward the steepness of the curve right at $x = 1$. (Both answers match the formula $2 + h$ from the worked example.)

2. From 8 am to 2 pm is 6 hours, so

   $$\frac{99.5 - 101.3}{6} = \frac{-1.8}{6} = -0.3 \ \text{°F per hour.}$$

   The fever fell, on average, by 0.3 degrees every hour. The units are degrees (output) per hour (input), and the minus sign says "falling".

3. Replace every x by $(x + h)$:

   $$\frac{3(x+h) + 2 - (3x + 2)}{h} = \frac{3x + 3h + 2 - 3x - 2}{h} = \frac{3h}{h} = 3.$$

The answer is 3 for every h, and does not even depend on x. A secant of a straight line is the line itself, so its slope never changes.

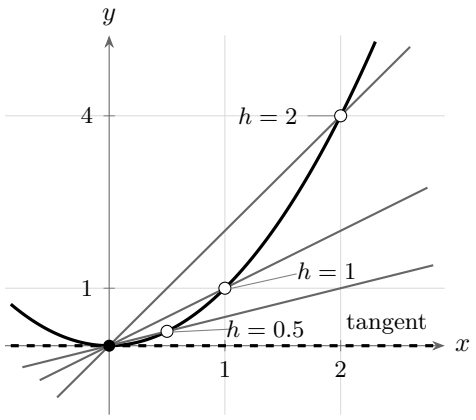
3 From Secant to Tangent

3.1 Shrink the Step

To keep the arithmetic as clean as possible, take $f(x) = x^2$ at the origin, $x = 0$. The difference quotient is

$$\frac{f(0+h) - f(0)}{h} = \frac{h^2 - 0}{h} = h \qquad (h \neq 0).$$

So the secant from 0 to h has slope exactly h. Now shrink the step:



step h	secant slope
2	2
1	1
0.5	0.5
0.1	0.1
0.01	0.01

Two things happen together as h shrinks:

- The second point, labelled by its h in the figure, slides down the curve toward $(0, 0)$.
- The secants flatten onto the dashed line, the x-axis, and their slopes home in on 0.

The dashed line is the **tangent line** at the origin: the line the secants are heading toward. Its slope, 0, is the steepness of the parabola right at its bottom.

Worked example: the car at $t = 1$. The same game at $x = 1$ uses the formula $2 + h$ from Section 2:

step h	1	0.5	0.1	0.01	0.001
secant slope $2 + h$	3	2.5	2.1	2.01	2.001

These are the car's average speeds over shorter and shorter intervals starting at $t = 1$. They are heading to 2, which suggests the speedometer reads 2 m/s at $t = 1$. Section 4 confirms it.

Notebook `U3-3_Derivatives-1_SecantToTangent.ipynb`, §2 and §3

3.2 The Problem with $h = 0$

Why not skip the shrinking and simply set $h = 0$? Because the difference quotient falls apart:

$$\frac{(0+0)^2 - 0^2}{0} = \frac{0}{0}.$$

Zero rise over zero run: this is the speedometer problem from Section 1, back again.

Why $0/0$ is not a number. Division undoes multiplication: $\frac{6}{2} = 3$ because $3 \cdot 2 = 6$, and 3 is the *only* number that works. If $\frac{0}{0}$ were some number c, we would need $c \cdot 0 = 0$. But that is true for **every** c: $5 \cdot 0 = 0$, $-17 \cdot 0 = 0$, $0.3 \cdot 0 = 0$. There is no single answer, so $0/0$ is **undefined**. It is not 0, not 1, not anything.

The way around it

- We may **not** set $h = 0$.
- We **may** make h as close to 0 as we like, and watch where the slope is heading.
- "Where it is heading" has a name: a **limit**.

3.3 Limits: Just Enough

Limits are a large subject, and a full course in calculus spends weeks on them. We need only the idea.

Definition (informal): limit

$$\lim_{h \to 0} g(h) = L$$

means that $g(h)$ gets **as close to L as we like** once h is close enough to 0, **without h ever equalling** 0. Read it aloud as "the limit of $g(h)$ as h goes to 0 is L".

From both sides. "Close to 0" includes negative h too: a step backward. For our quotient $g(h) = h$ at the origin:

h	slope	h	slope
0.1	0.1	-0.1	-0.1
0.01	0.01	-0.01	-0.01
0.001	0.001	-0.001	-0.001

A negative h means the second point sits to the left of the origin, so the secant tilts down instead of up. Both sides head to the same place, so the limit is 0.

What a limit ignores. The value **at** $h = 0$ itself plays no part. That is the whole point: the quotient has *no* value at $h = 0$, yet it still heads somewhere, and the limit reports where.

A non-example: the two sides disagree. For $f(x) = |x|$ (absolute value) at $x = 0$, the difference quotient is $\frac{|h|}{h}$. For every positive h this is 1; for every negative h it is -1. The right side heads to 1 and the left side to -1, so there is **no** single place it heads, and the limit does not

exist. In the picture, the graph of $|x|$ has a sharp corner at the origin, and no one line can claim to be the tangent there. We will not meet such corners again in this unit; every function we use is smooth.

Notebook `U3-2_Limits-1_ApproachingALimit.ipynb`, §1 and §3

3.4 Computing a Limit: Simplify, Then Shrink

Tables suggest a limit; algebra proves it. The method is always the same two steps.

Worked example: the slope of x^2 at $x = 0$.

$$\begin{aligned} \frac{(0+h)^2 - 0^2}{h} &= \frac{h^2}{h} && \text{expand} \\ &= \frac{h \cdot h}{h} && \text{factor out } h \\ &= h && \text{cancel, allowed since } h \neq 0 \\ \lim_{h \to 0} h &= 0 && \text{now let } h \text{ shrink} \end{aligned}$$

The move

- The h that caused $0/0$ **cancels**. After that, nothing stops h from going to 0: we can simply put $h = 0$ into what is left.
- Every limit in this unit goes this way: **simplify first, then let h shrink**.

The same steps work at any point. Here are three more.

Worked example: x^2 at $x = 1$. Expand $(1+h)^2 = 1 + 2h + h^2$:

$$\frac{(1+h)^2 - 1}{h} = \frac{2h + h^2}{h} = \frac{h\,(2+h)}{h} = 2 + h \xrightarrow{h \to 0} 2.$$

The table guessed 2; now we know it.

Worked example: x^2 at $x = 3$. Expand $(3+h)^2 = 9 + 6h + h^2$:

$$\frac{(3+h)^2 - 9}{h} = \frac{6h + h^2}{h} = 6 + h \xrightarrow{h \to 0} 6.$$

Worked example: x^3 at $x = 1$. Expand $(1+h)^3 = 1 + 3h + 3h^2 + h^3$:

$$\frac{(1+h)^3 - 1}{h} = \frac{3h + 3h^2 + h^3}{h} = 3 + 3h + h^2 \xrightarrow{h \to 0} 3.$$

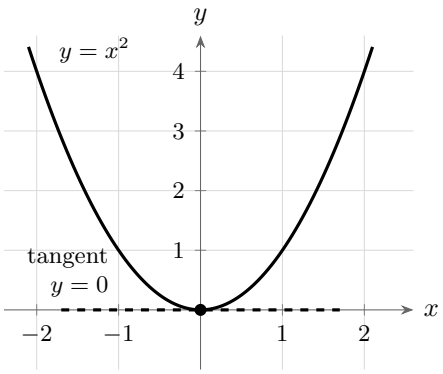
Notice the slopes of x^2: 0 at $x = 0$, 2 at $x = 1$, 6 at $x = 3$. Each is twice the point. Section 4 explains why.

Watch out

- **Do not plug in $h = 0$ first.** You will get $0/0$ every time; that is the problem we are solving, not the answer.

- **Cancel factors, not terms.** You may cancel h only when it divides *every* term of the top. $\frac{2h+h^2}{h} = 2+h$ is right; $\frac{2h+h^2}{h} = 2h+h$ (cancelling only from the last term) is wrong.

3.5 The Tangent Line



Definition: tangent line

The **tangent line** to a graph at a point is the line the secants approach as $h \to 0$.

- It touches the graph at that **one** point.
- It runs in the **same direction** as the curve there: zoom in far enough and the curve and its tangent look the same.
- Its slope is the **instantaneous** rate of change.

For $y = x^2$ at the origin, the tangent is the flat line $y = 0$, slope 0.

Watch out: "touches at one point" is about that point

A tangent touches the graph at one point *near the point of tangency*; far away it may cross the graph again. For example, the tangent to $y = x^3 - 3x$ at $x = -1$ is the flat line $y = 2$, which meets the curve again at $x = 2$. It is still the tangent at $x = -1$.

Back to the car. At $t = 0$ the tangent to $x(t) = t^2$ is flat, so the speedometer reads 0 m/s: the car at rest, just as it pulls away. The reading at $t = 1$, which the table guessed as 2 m/s, is confirmed in the next section.

Notebook `U3-3_Derivatives-1_SecantToTangent.ipynb`, §4

Exercise 3.1

Use "simplify, then shrink" to find each slope.

1. The slope of $y = x^2$ at $x = -2$.
2. The slope of $y = 4x - 7$ at $x = 10$.
3. The slope of $y = x^2 + 5$ at $x = 1$. Compare with the slope of x^2 at $x = 1$; why are they the same?

Solution to Exercise 3.1

1. Expand $(-2+h)^2 = 4 - 4h + h^2$:

$$\frac{(-2+h)^2 - 4}{h} = \frac{-4h + h^2}{h} = -4 + h \;\to\; -4.$$

   Negative, as it should be: left of the bottom, the parabola is falling. (And again, twice the point.)

2. $f(10+h) = 4(10+h) - 7 = 33 + 4h$ and $f(10) = 33$:

$$\frac{33 + 4h - 33}{h} = \frac{4h}{h} = 4 \;\to\; 4.$$

   A line's slope is the same everywhere.

3. $f(1+h) = (1+h)^2 + 5 = 6 + 2h + h^2$ and $f(1) = 6$:

$$\frac{6 + 2h + h^2 - 6}{h} = \frac{2h + h^2}{h} = 2 + h \;\to\; 2.$$

   The same as for x^2. Adding 5 lifts the whole graph up without tilting it, so every slope is unchanged; the 5 cancels in the subtraction.

4 The Derivative

4.1 The Definition

Everything so far was at one point at a time. The derivative does it at **every** point at once.

Definition: derivative

The **derivative** of f at x is the limit of the difference quotient:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

- It is the slope of the **tangent line** at x.
- It is the **instantaneous rate of change** of f at x, in output units **per** input unit: for a drug level over time, mg/L per hour.
- To **differentiate** f is to find its derivative. If the limit exists at x, f is **differentiable** at x.

Because x is left as a letter, the answer $f'(x)$ is a new **function**: put in any x and it returns the slope there.

	secant	**tangent**
points on the graph	two	one
step	$h \neq 0$	$h \to 0$
slope means	average rate of change	instantaneous rate of change

4.2 Notation

The same derivative has several names, because it was invented twice (by Newton and by Leibniz) and different fields kept different habits.

$f'(x)$	"f prime of x". Short, and good for a function with a name.
$\dfrac{df}{dx}$	"df dx": a tiny rise df over a tiny run dx, the limit of $\Delta f/\Delta x$. It shows what changes with respect to what.
$\dfrac{d}{dx}f(x)$	an **operation** applied to f: in goes a function, out comes its derivative. Handy for a formula with no name, as in $\frac{d}{dx}(x^3+1)$.
$\dot{x}(t)$	"x dot": used only for rates **with respect to time**, $\dot{x} = \dfrac{dx}{dt}$. Common in physics and in models of drugs in the body.

If the output is called y, as in $y = f(x)$, you will also see y' and $\dfrac{dy}{dx}$. The car's velocity is $\dot{x}(t)$, and a drug concentration's rate of change is $\dfrac{dC}{dt}$ or $\dot{C}(t)$.

To report the slope at one particular input, find the derivative *first* and then put the number in: $f'(3)$ means "the derivative function, evaluated at 3".

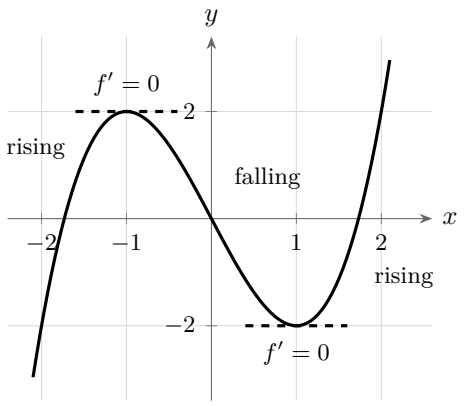
Watch out

- $f'(3)$ is **not** the derivative of the number $f(3)$. The number $f(3)$ is a constant, and its derivative would be 0. Differentiate, then substitute.
- In $\frac{dy}{dx}$, the d's are not variables to cancel. Read the whole thing as one symbol.

Several inputs: a preview. When f has more than one input, as in $f(x, y)$, the derivative with respect to x is written $\dfrac{\partial f}{\partial x}$, "partial f, partial x". The curly ∂ is a signal that f has **other** inputs, held fixed while only x moves. Section 7 builds this.

4.3 Reading a Derivative off a Graph

The derivative can be read from a graph without any algebra, by looking at the slope of the tangent at each point.



$y = x^3 - 3x$

$f'(x) > 0$ graph **rising**

$f'(x) < 0$ graph **falling**

$f'(x) = 0$ graph **flat**

Reading the graph of $y = x^3 - 3x$ from left to right:

- Left of $x = -1$ the graph climbs, so $f' > 0$ there.
- At $x = -1$ it reaches a peak, and the tangent is flat: $f'(-1) = 0$.
- Between -1 and 1 it falls, so $f' < 0$.
- At $x = 1$ it bottoms out in a valley: $f'(1) = 0$.
- Right of $x = 1$ it climbs again: $f' > 0$.

The **size** of f' tells steepness as well: the graph is steepest, falling fastest, as it crosses the origin. Section 5 will show that this function's derivative is $f'(x) = 3x^2 - 3$, which is indeed 0 at $x = \pm 1$, negative between them, positive outside, and most negative (-3) at $x = 0$.

Why we care. The flat spots are the peaks and valleys. Fitting a model means finding the lowest valley of its error, and Unit 3-2 hunts for exactly these points by solving $f'(x) = 0$.

4.4 Six Derivatives from the Definition

Each example below uses the same three steps: write the difference quotient, simplify until the h in the denominator cancels, then let $h \to 0$.

Example: $f(x) = c$, a constant

Here $f(x + h) = c$ too, because there is no x in the formula to replace.

$$f'(x) = \lim_{h \to 0} \frac{c - c}{h} = \lim_{h \to 0} \frac{0}{h} = 0.$$

A constant never changes, so its rate of change is 0 everywhere. Its graph is a flat horizontal line, and the tangent at every point is the line itself. Note that $\frac{0}{h} = 0$ because $h \neq 0$; the limit never divides by zero.

Example: $f(x) = x$

Here $f(x+h) = x + h$:

$$f'(x) = \lim_{h \to 0} \frac{(x+h) - x}{h} = \lim_{h \to 0} \frac{h}{h} = 1.$$

The line $y = x$ rises 1 for every 1 across, everywhere.

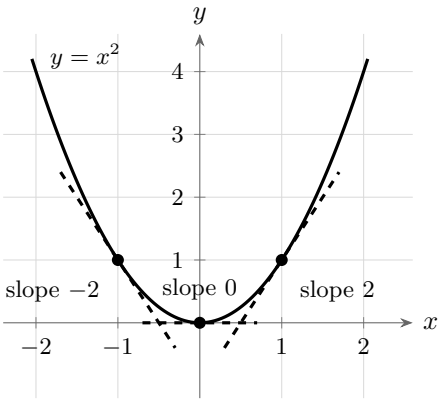
Example: $f(x) = mx + b$**, any line**

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{m(x+h) + b - (mx + b)}{h} \\ &= \lim_{h \to 0} \frac{mx + mh + b - mx - b}{h} \\ &= \lim_{h \to 0} \frac{mh}{h} = m. \end{aligned}$$

A line's derivative is its **slope** m, the same at every x. The intercept b slides the line up or down without tilting it, so it drops out. The two previous examples were special cases: a constant is the line with $m = 0$, and $y = x$ is the line with $m = 1$ and $b = 0$. For example, $y = 2x + 1$ has derivative 2 everywhere.

Example: $f(x) = x^2$

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h} \\ &= \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \\ &= \lim_{h \to 0} \frac{h\,(2x + h)}{h} \\ &= \lim_{h \to 0} (2x + h) = 2x. \end{aligned}$$



This is the pattern noticed in Section 3: the slope of x^2 is twice the point. The single formula $f'(x) = 2x$ gives the slope at **every** x at once: -2 at $x = -1$, 0 at $x = 0$, then $2, 4, 6$ at $x = 1, 2, 3$. In particular, the car's speedometer at $t = 1$ reads $\dot{x}(1) = 2 \cdot 1 = 2$ m/s, as the table predicted.

Example: $f(x) = \sin x$

This one needs a fact from trigonometry, the angle-addition identity

$$\sin(x+h) = \sin x \cos h + \cos x \sin h.$$

Substituting and regrouping splits the quotient into two pieces:

$$\begin{aligned}\frac{\sin(x+h) - \sin x}{h} &= \frac{\sin x \cos h + \cos x \sin h - \sin x}{h} \\ &= \sin x \cdot \frac{\cos h - 1}{h} + \cos x \cdot \frac{\sin h}{h}.\end{aligned}$$

Now $\sin x$ and $\cos x$ do not involve h, so only the two fractions change as $h \to 0$. Their limits can be read off a table (with h in radians):

h	$\dfrac{\sin h}{h}$	$\dfrac{\cos h - 1}{h}$
0.1	0.9983	-0.0500
0.01	0.99998	-0.0050
$\to 0$	$\to 1$	$\to 0$

So

$$f'(x) = \sin x \cdot 0 + \cos x \cdot 1 = \cos x.$$

A check against the graph. The sine curve has slope 1 as it rises through $x = 0$, slope 0 at its peak at $x = \frac{\pi}{2}$, slope -1 as it falls through $x = \pi$, and slope 0 at its trough at $x = \frac{3\pi}{2}$. Those four slopes are exactly $\cos x$ at the same four points. This works only when x is measured in **radians**; in degrees the table's limits come out differently and the formula picks up an extra constant.

Example: $f(x) = 1/x$

The new step here is combining two fractions in the numerator over a common denominator:

$$\frac{1}{x+h} - \frac{1}{x} = \frac{x}{x\,(x+h)} - \frac{x+h}{x\,(x+h)} = \frac{x - (x+h)}{x\,(x+h)} = \frac{-h}{x\,(x+h)}.$$

Now divide by h, which cancels:

$$\frac{f(x+h) - f(x)}{h} = \frac{-h}{x\,(x+h)} \cdot \frac{1}{h} = \frac{-1}{x\,(x+h)}.$$

Let $h \to 0$:

$$f'(x) = -\frac{1}{x \cdot x} = -\frac{1}{x^2} \qquad (x \neq 0).$$

Since $x^2 > 0$, the derivative is **always negative**: $1/x$ is always falling. Its size tells how steeply: at $x = \frac{1}{2}$ the slope is -4, at $x = 1$ it is -1, at $x = 2$ only $-\frac{1}{4}$. Steep near 0, nearly flat far out.

One more: $f(x) = x^3$

The pattern from x and x^2 continues. Expand $(x+h)^3 = x^3 + 3x^2h + 3xh^2 + h^3$:

$$f'(x) = \lim_{h\to 0} \frac{3x^2h + 3xh^2 + h^3}{h} = \lim_{h\to 0} \left(3x^2 + 3xh + h^2\right) = 3x^2.$$

So far: $x \to 1,\ x^2 \to 2x,\ x^3 \to 3x^2$. The power comes down in front, and the new power is one less. Section 5 turns this into a rule.

Exercise 4.1

1. Use the limit definition to find the derivative of $f(x) = 3x^2$.
2. Use the limit definition to find the derivative of $f(x) = x^2 + x$.
3. A drug's blood concentration has $\dfrac{dC}{dt} = -1.5$ mg/L per hour at $t = 4$ hours. In words, what does that mean?

Solution to Exercise 4.1

1. Expand $3(x+h)^2 = 3x^2 + 6xh + 3h^2$:

$$\frac{3(x+h)^2 - 3x^2}{h} = \frac{6xh + 3h^2}{h} = \frac{h\,(6x + 3h)}{h} = 6x + 3h \ \to\ 6x.$$

   That is 3 times the derivative of x^2: the constant just rides along.

2. $f(x+h) = (x+h)^2 + (x+h) = x^2 + 2xh + h^2 + x + h$, so

$$\frac{x^2 + 2xh + h^2 + x + h - x^2 - x}{h} = \frac{2xh + h^2 + h}{h} = 2x + h + 1 \ \to\ 2x + 1.$$

   That is the derivative of x^2 plus the derivative of x: the sum splits.

3. At 4 hours the concentration is **falling** (the sign is negative), at a rate of 1.5 mg/L per hour. If that pace held for one more hour, it would drop by about 1.5 mg/L. It is an instantaneous rate, so the real drop over the next hour may differ, since the rate itself changes.

Hold on to answers 1 and 2: they are two of the rules in the next section.

5 Derivative Rules

Taking a limit every time is slow. The rules below are shortcuts, each proved once from the definition and then used forever.

5.1 The Power Rule

Power rule

$$\frac{d}{dx}\,x^n = n\,x^{n-1} \qquad \text{for any number } n.$$

In words: bring the power down in front, then lower the power by one.

Section 4 checked it for $n = 1, 2, 3$, and for $n = -1$ too, since $1/x = x^{-1}$ and $-1 \cdot x^{-2} = -\frac{1}{x^2}$. It holds for every n: negative, fractional, anything.

function		as a power		derivative		simplified
x^3			$\longrightarrow$	$3x^2$		
x^{10}			$\longrightarrow$	$10x^9$		
$\sqrt{x}$	$=$	$x^{1/2}$	$\longrightarrow$	$\frac{1}{2}x^{-1/2}$	$=$	$\dfrac{1}{2\sqrt{x}}$
$\dfrac{1}{x^2}$	$=$	x^{-2}	$\longrightarrow$	$-2x^{-3}$	$=$	$-\dfrac{2}{x^3}$
$\sqrt[3]{x}$	$=$	$x^{1/3}$	$\longrightarrow$	$\frac{1}{3}x^{-2/3}$	$=$	$\dfrac{1}{3\sqrt[3]{x^2}}$
$\dfrac{1}{\sqrt{x}}$	$=$	$x^{-1/2}$	$\longrightarrow$	$-\frac{1}{2}x^{-3/2}$	$=$	$-\dfrac{1}{2x\sqrt{x}}$

Two quick consistency checks: $x^1 \to 1 \cdot x^0 = 1$, matching the derivative of x; and $x^0 = 1$ (a constant) $\to 0 \cdot x^{-1} = 0$, matching the derivative of a constant.

Watch out

- **Rewrite roots and fractions as powers first.** A root is a fractional power ($\sqrt{x} = x^{1/2}$), and x in a denominator is a negative power ($\frac{1}{x^3} = x^{-3}$).
- **Lowering a negative power makes it more negative**: $-2 - 1 = -3$, not -1.
- **The power rule needs the variable in the base and a constant in the exponent.** 2^x is not a power of x, and its derivative is *not* $x \cdot 2^{x-1}$. It is in the table in Section 5.3.

Notebook `U3-3_Derivatives-2_Rules.ipynb`, §2

5.2 Constant Multiple and Sum Rules

Constant multiple rule

$$\frac{d}{dx}\big[c\,f(x)\big] = c\,f'(x).$$

Constants ride along. (Exercise 4.1, part 1: $3x^2 \to 3 \cdot 2x$.)

Sum rule

$$\frac{d}{dx}\big[f(x) \pm g(x)\big] = f'(x) \pm g'(x).$$

Sums and differences split. (Exercise 4.1, part 2: $x^2 + x \to 2x + 1$.)

Together with the power rule, these differentiate any polynomial term by term.

Worked example.

$$\frac{d}{dx}(4x^3 - 2x + 7) = 4 \cdot 3x^2 \;-\; 2 \cdot 1 \;+\; 0 = 12x^2 - 2.$$

The power rule on each term, constants carried, and the lone 7 (a constant) differentiates to 0.

Worked example.

$$\frac{d}{dx}(6x^5 - 4x^3 + x - 9) = 30x^4 - 12x^2 + 1.$$

Worked example: powers in disguise. For $f(x) = \dfrac{3}{x^2} + 4\sqrt{x}$, rewrite first: $f(x) = 3x^{-2} + 4x^{1/2}$. Then

$$f'(x) = 3 \cdot (-2)x^{-3} + 4 \cdot \tfrac{1}{2}x^{-1/2} = -\frac{6}{x^3} + \frac{2}{\sqrt{x}}.$$

Worked example: divide first. For $f(x) = \dfrac{x^2 + 1}{x}$, split the fraction term by term: $f(x) = \dfrac{x^2}{x} + \dfrac{1}{x} = x + x^{-1}$. Then $f'(x) = 1 - x^{-2} = 1 - \dfrac{1}{x^2}$. (The quotient rule below gives the same answer with more work.)

Worked example: back to Section 4. $f(x) = x^3 - 3x$ has $f'(x) = 3x^2 - 3$, as promised.

Watch out

A constant **added** differentiates to 0, but a constant **multiplying** stays. In $5x^2 + 5$, the first 5 survives ($10x$) and the second vanishes.

5.3 Common Derivatives

These come up constantly; each can be proved from the definition, as $\sin x$ was.

$f(x)$	$f'(x)$
c	0
x^n	$n\,x^{n-1}$
e^x	e^x
$\ln x$	$\dfrac{1}{x}\quad (x>0)$
a^x	$a^x \ln a$
$\log_a x$	$\dfrac{1}{x \ln a}\quad (x>0)$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$

- e^x is its **own derivative**: at every point, its slope equals its height. That is what makes e the natural base for growth and decay.
- $\ln x$ is the logarithm with base e. Its derivative $\frac{1}{x}$ is positive (always rising) but shrinks: $\ln x$ keeps growing, ever more slowly.
- The rows for a general base $a > 0$ reduce to the base-e rows when $a = e$, because $\ln e = 1$: $e^x \ln e = e^x$ and $\frac{1}{x \ln e} = \frac{1}{x}$. A base other than e just picks up the constant $\ln a$.
- The trigonometric rows need x in radians.

Worked examples.

$$\frac{d}{dx}\,2^x = 2^x \ln 2 \approx 0.693 \cdot 2^x,$$
$$\frac{d}{dx}\,\log_{10} x = \frac{1}{x \ln 10},$$
$$\frac{d}{dx}\left(4e^x + 3\ln x\right) = 4e^x + \frac{3}{x},$$
$$\frac{d}{dx}\left(5\sin x - 2\cos x\right) = 5\cos x - 2(-\sin x) = 5\cos x + 2\sin x.$$

The last one is a sign trap: the derivative of $\cos x$ is $-\sin x$, and subtracting it turns into adding.

5.4 The Product Rule

Product rule

$$\frac{d}{dx}\left[f(x)\,g(x)\right] = f'(x)\,g(x) + f(x)\,g'(x)$$

In words: (first)′ × second + first × (second)′.

Each factor takes a turn being differentiated while the other stays as it is, and the two results add.

Worked example: $x^2 e^x$**.**

$$\begin{aligned} & f = x^2, \quad g = e^x && \text{name the first and the second} \\ & f' = 2x, \quad g' = e^x && \text{differentiate each one} \\ \frac{d}{dx}\left[x^2 e^x\right] \; & = 2x \cdot e^x + x^2 \cdot e^x && \text{assemble } f'g + fg' \\ & = x\, e^x\,(2 + x) && \text{factor out } x\, e^x \end{aligned}$$

Worked example: $x \sin x$**.** With $f = x$, $g = \sin x$: $f' = 1$, $g' = \cos x$, so

$$\frac{d}{dx}\left[x \sin x\right] = 1 \cdot \sin x + x \cos x = \sin x + x \cos x.$$

Worked example: $e^x \ln x$**.** With $f = e^x$, $g = \ln x$:

$$\frac{d}{dx}\left[e^x \ln x\right] = e^x \ln x + e^x \cdot \frac{1}{x} = e^x \left(\ln x + \frac{1}{x}\right).$$

A check we can verify another way. $x^2 \cdot x^3 = x^5$, whose derivative is $5x^4$ by the power rule. The product rule gives $2x \cdot x^3 + x^2 \cdot 3x^2 = 2x^4 + 3x^4 = 5x^4$. The same.

Watch out

The derivative of a product is not the product of the derivatives. Take $x \cdot x = x^2$: the product of the derivatives would be $1 \cdot 1 = 1$, but the true derivative is $2x$. Both terms of the product rule are needed.

Notebook `U3-3_Derivatives-2_Rules.ipynb`, §3

5.5 The Quotient Rule

Quotient rule

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)\, g(x) - f(x)\, g'(x)}{g(x)^2}$$

In words: "low d-high minus high d-low, over low squared" (low = bottom, high = top, d = derivative of).

Worked example: $x/(x+1)$**.**

$$\begin{aligned} & f = x, \quad g = x + 1 && \text{name the top and the bottom} \\ & f' = 1, \quad g' = 1 && \text{differentiate each one} \\ \frac{d}{dx}\left[\frac{x}{x+1}\right] \; & = \frac{1 \cdot (x+1) - x \cdot 1}{(x+1)^2} && \text{assemble } \frac{f'g - fg'}{g^2} \\ & = \frac{x + 1 - x}{(x+1)^2} = \frac{1}{(x+1)^2} && \text{simplify the top} \end{aligned}$$

The answer is always positive, so $\frac{x}{x+1}$ is always rising (where it is defined), more and more slowly.

Worked example: $\dfrac{\ln x}{x}$**.** Top $f = \ln x$, $f' = \frac{1}{x}$; bottom $g = x$, $g' = 1$:

$$\frac{\frac{1}{x} \cdot x - \ln x \cdot 1}{x^2} = \frac{1 - \ln x}{x^2}.$$

Worked example: $\dfrac{x^2}{x-1}$**.** Top $f = x^2$, $f' = 2x$; bottom $g = x - 1$, $g' = 1$:

$$\frac{2x(x-1) - x^2 \cdot 1}{(x-1)^2} = \frac{2x^2 - 2x - x^2}{(x-1)^2} = \frac{x^2 - 2x}{(x-1)^2}.$$

A check. $\frac{1}{x}$ as a quotient, top 1 and bottom x: $\frac{0 \cdot x - 1 \cdot 1}{x^2} = -\frac{1}{x^2}$, matching Section 4.

Watch out

- **Order matters in the top.** Because of the minus sign, $f'g - fg'$ and $fg' - f'g$ differ by a sign. The mnemonic starts with "low": the bottom, times the derivative of the top.
- **Do not forget to square the bottom.**
- **A constant bottom needs no quotient rule.** $\frac{x^3}{5} = \frac{1}{5}x^3$, whose derivative is $\frac{3}{5}x^2$ by the constant multiple rule.

Exercise 5.1

Differentiate.

1. $5x^4 - 3x^2 + 8$
2. $2\sqrt{x} + \dfrac{1}{x}$
3. $3e^x - 4\ln x$
4. $x^3 \ln x$
5. $\dfrac{e^x}{x}$
6. $(x+1)^2$ (expand it first)

Solution to Exercise 5.1

1. Term by term, with the power rule and constants carried: $5 \cdot 4x^3 - 3 \cdot 2x + 0 = 20x^3 - 6x$.
2. Rewrite as powers, $2x^{1/2} + x^{-1}$. Then
$$2 \cdot \tfrac{1}{2}x^{-1/2} + (-1)x^{-2} = \frac{1}{\sqrt{x}} - \frac{1}{x^2}.$$
3. From the table, with constants carried: $3e^x - 4 \cdot \dfrac{1}{x} = 3e^x - \dfrac{4}{x}$.
4. A product, with $f = x^3$ ($f' = 3x^2$) and $g = \ln x$ ($g' = \frac{1}{x}$):
$$3x^2 \ln x + x^3 \cdot \frac{1}{x} = 3x^2 \ln x + x^2 = x^2(3\ln x + 1).$$
5. A quotient, with top e^x (derivative e^x) and bottom x (derivative 1):
$$\frac{e^x \cdot x - e^x \cdot 1}{x^2} = \frac{e^x(x-1)}{x^2}.$$

6. $(x+1)^2 = x^2 + 2x + 1$, so the derivative is $2x + 2$. Expanding would not be practical for $(x+1)^{50}$: the chain rule, next, does it without expanding.

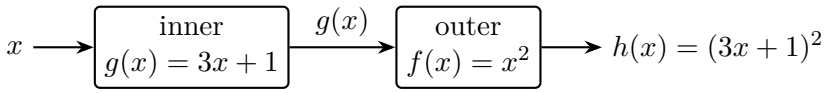
6 The Chain Rule

6.1 Functions Inside Functions

Unit 1-3 introduced **composition**: feeding the output of one function into another.

$$h(x) = (f \circ g)(x) = f(g(x)).$$

The **inner** function g acts first, and the **outer** function f acts on its output.



For $x = 2$: the inner function gives $g(2) = 7$, and the outer squares it, $f(7) = 49$. So $h(2) = (3 \cdot 2 + 1)^2 = 49$.

Reading a composition backward. To use the chain rule we must take a formula apart into its inner and outer pieces. Ask: *if I computed this by hand, what would I do last?* That last step is the outer function, and whatever it acts on is the inner function.

function h	inner g	outer f	last step by hand
$e^{-0.2t}$	$-0.2t$	e^t	raise e to the power
$\sqrt{x^2+1}$	x^2+1	$\sqrt{x}$	take the square root
$(5-2w)^2$	$5-2w$	w^2	square
$\sin(2x)$	$2x$	$\sin x$	take the sine
$\ln(x^2+1)$	x^2+1	$\ln x$	take the log

The letter used for the outer function's input does not matter: the outer function in the first row is "e to the power of its input", whatever we call that input.

Watch out: order matters

$f(g(x))$ and $g(f(x))$ are different functions. With $g(x) = 3x + 1$ and $f(x) = x^2$: $f(g(x)) = (3x+1)^2$, but $g(f(x)) = 3x^2 + 1$.

Why we care. A model's error on one data point is often the **square** of (actual minus predicted), like the third row. Its inner function is "actual minus predicted", its outer function is "square". Unit 3-2 differentiates exactly this.

6.2 The Chain Rule

Chain rule

$$h(x) = f(g(x)) \quad \Longrightarrow \quad h'(x) = f'(g(x)) \cdot g'(x)$$

In words: the derivative of the outer, at the inner, times the derivative of the inner.

Why multiply? Rates multiply. If g changes 3 times as fast as x, and f changes 2 times as fast as g, then $f(g(x))$ changes $3 \cdot 2 = 6$ times as fast as x.

Worked example: $h(x) = (3x+1)^2$**.**

$$\begin{aligned} & g(x) = 3x+1, \quad f(x) = x^2 && \text{name the inner and the outer} \\ & g'(x) = 3, \quad f'(x) = 2x && \text{differentiate each one} \\ & f'(g(x)) = 2(3x+1) && \text{outer's derivative, at the inner} \\ h'(x) \quad & = 2(3x+1) \cdot 3 = 18x + 6 && \text{times } g'(x) \end{aligned}$$

Check by expanding: $(3x+1)^2 = 9x^2 + 6x + 1$, whose derivative is $18x + 6$. The same, but the chain rule never needed the expansion, which matters for $(3x+1)^{50}$: its derivative is $50(3x+1)^{49} \cdot 3 = 150(3x+1)^{49}$ in one line.

Worked example: a drug washing out, $C(t) = 10\,e^{-0.2t}$ **mg/L.** Inner $g(t) = -0.2t$ with $g'(t) = -0.2$; outer "e to the power", which is its own derivative. With the constant 10 carried along,

$$C'(t) = 10 \cdot e^{-0.2t} \cdot (-0.2) = -2\,e^{-0.2t} \text{ mg/L per hour.}$$

At $t = 0$ the concentration falls at 2 mg/L per hour. Notice that $C'(t) = -0.2\,C(t)$: the drug is cleared at a rate proportional to how much is there, which is why a high level falls fast and a low level falls slowly.

Worked example: an error, $E(w) = (5-2w)^2$**.** Inner $g(w) = 5 - 2w$ with $g'(w) = -2$; outer "square", derivative "twice the input":

$$E'(w) = 2(5-2w) \cdot (-2) = -4(5-2w).$$

At $w = 1$, $E'(1) = -4 \cdot 3 = -12$. Negative means the error is falling as w increases, so a slightly larger w gives a smaller error. This kind of reading is the heart of Unit 3-2.

Worked example: $\sin(2x)$**.** Inner $2x$ (derivative 2), outer sin (derivative cos):

$$\frac{d}{dx}\sin(2x) = \cos(2x) \cdot 2 = 2\cos(2x).$$

Worked example: $\ln(x^2+1)$**.** Inner x^2+1 (derivative $2x$), outer ln (derivative $\frac{1}{\text{input}}$):

$$\frac{d}{dx}\ln(x^2+1) = \frac{1}{x^2+1} \cdot 2x = \frac{2x}{x^2+1}.$$

Watch out

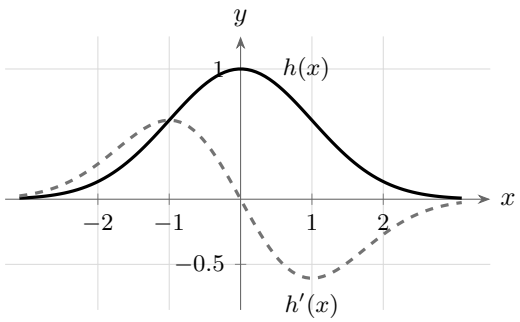
- **Do not forget the inner derivative.** The most common chain-rule mistake is writing $\frac{d}{dx}(3x+1)^2 = 2(3x+1)$ and stopping. That misses the factor 3.
- **Evaluate the outer derivative at the inner function, not at x.** For $(3x+1)^2$ the outer derivative is $2 \cdot (\text{input})$, and the input is $3x+1$: so $2(3x+1)$, not $2x$.
- **When the inner function is just x, there is nothing to do.** The chain rule's extra factor is then 1, which is why the earlier rules never needed it.

Notebook `U3-3_Derivatives-2_Rules.ipynb`, §4

6.3 The Chain Rule: the Bell Curve

The function $h(x) = e^{-x^2/2}$ gives the shape of the normal distribution, the bell curve of statistics (Unit 4).

$$
\begin{aligned}
& g(x) = -\tfrac{1}{2}x^2, \quad f(x) = e^x && \text{name the inner and the outer} \\
& g'(x) = -x, \quad f'(x) = e^x && \text{differentiate each one} \\
h'(x) \; & = e^{-x^2/2} \cdot (-x) = -x\, e^{-x^2/2} && \text{assemble } f'(g(x)) \cdot g'(x)
\end{aligned}
$$



What the derivative says, read against the curve:

- Left of 0, $x < 0$ so $-x > 0$ and $h' > 0$: the bell is **rising**.
- Right of 0, $h' < 0$: the bell is **falling**.
- At $x = 0$, $h'(0) = 0$: the flat top of the bell, its **peak**.
- The bell is steepest at $x = \pm 1$, one standard deviation from the center. Some values of the slope on the right side: $h'(0.5) \approx -0.44$, $h'(1) \approx -0.61$, $h'(2) \approx -0.27$. The fall is fastest at $x = 1$. (Proving that $x = 1$ is exactly the steepest point uses the tools of Unit 3-2.)

Exercise 6.1

Differentiate.

1. $(2x-5)^3$
2. e^{3x}

3. $\sqrt{x^2+1}$

Solution to Exercise 6.1

1. Inner $g(x) = 2x - 5$, with $g'(x) = 2$. Outer $f(x) = x^3$, with $f'(x) = 3x^2$. The outer derivative at the inner is $3(2x-5)^2$, so

$$\frac{d}{dx}(2x-5)^3 = 3(2x-5)^2 \cdot 2 = 6(2x-5)^2.$$

2. Inner $g(x) = 3x$, with $g'(x) = 3$. Outer $f(x) = e^x$, its own derivative, so at the inner it is e^{3x}:

$$\frac{d}{dx}e^{3x} = e^{3x} \cdot 3 = 3e^{3x}.$$

3. Inner $g(x) = x^2 + 1$, with $g'(x) = 2x$. Outer $f(x) = \sqrt{x} = x^{1/2}$, with $f'(x) = \frac{1}{2}x^{-1/2}$. At the inner, $\frac{1}{2}(x^2+1)^{-1/2}$, so

$$\frac{d}{dx}\sqrt{x^2+1} = \tfrac{1}{2}(x^2+1)^{-1/2} \cdot 2x = \frac{x}{\sqrt{x^2+1}}.$$

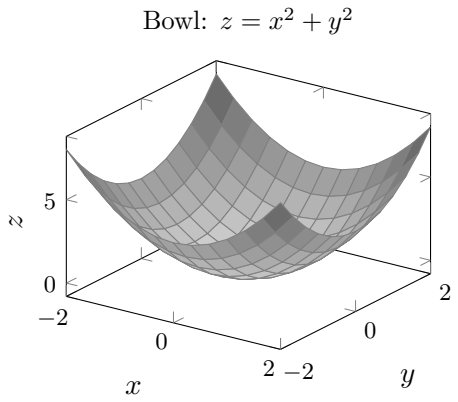
   The 2 from the inner derivative cancels the $\frac{1}{2}$ from the power rule.

7 Partial Derivatives

7.1 Functions of Two Variables, from Unit 1-3

A function $f(x, y)$ of two variables takes a **pair** of numbers and returns **one**. Its graph is a **surface**: each point (x, y) on the floor gets a height $z = f(x, y)$ above it.

body mass index	$B(w, h) = w/h^2$
drug concentration	$C(D, V) = D/V$
a bowl	$f(x, y) = x^2 + y^2$
a model's loss	$L(m, b)$



In the examples: BMI depends on weight w (kg) and height h (m); a drug's concentration depends on the dose D (mg) and the volume V (L) it spreads through; and a straight-line model's loss depends on its slope m and intercept b, which Unit 3-2 builds. The bowl has one lowest point, at $(0, 0)$, and finding the lowest point of a loss surface is exactly what training a model means.

Notebook `U3-6_Multivariable-1_Surface3D.ipynb`, §1 and §2

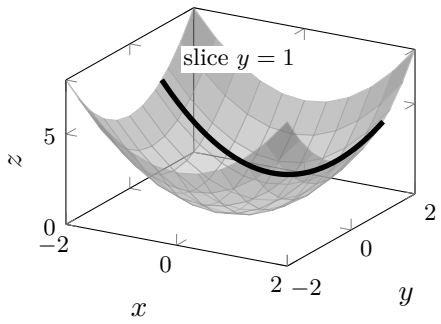
7.2 Freeze One Variable

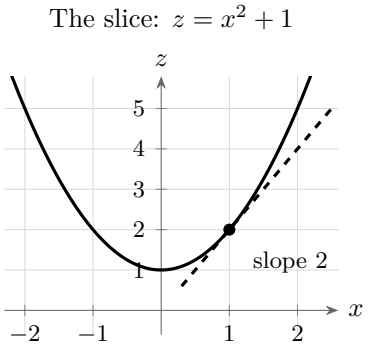
A surface has no single slope at a point: walking east and walking north can climb at different rates. So we ask about one direction at a time.

The idea, for $f(x, y) = x^2 + y^2$. Hold $y = 1$ fixed. Then

$$f(x, 1) = x^2 + 1$$

is a function of x **alone**. Its graph is the curve we get by slicing the bowl with the wall $y = 1$: a parabola.





The slice is an ordinary one-variable function, so the rules from earlier apply: its derivative is $2x$, which is 2 at $x = 1$ (the dashed tangent on the right). That number is the **partial derivative of f with respect to x** at the point $(1, 1)$: the slope of the bowl at $(1, 1)$ if you walk in the x direction.

The other direction works the same way. Hold $x = 1$ fixed: $f(1, y) = 1 + y^2$, with derivative $2y$, which is 2 at $y = 1$. That is the partial derivative with respect to y at $(1, 1)$. (They are equal here only because the bowl is symmetric.)

Notebook `U3-6_Multivariable-2_PartialDerivatives.ipynb`, §1 and §3

7.3 The Definition

Definition: partial derivatives

$$\frac{\partial f}{\partial x} = \lim_{h \to 0} \frac{f(x + h,\, y) - f(x, y)}{h} \qquad \text{only } x \text{ moves}$$

$$\frac{\partial f}{\partial y} = \lim_{h \to 0} \frac{f(x,\, y + h) - f(x, y)}{h} \qquad \text{only } y \text{ moves}$$

The same difference quotient as before, with the other variable frozen.

Notation. The symbol ∂ is a curly d, read "partial": $\frac{\partial f}{\partial x}$ is read "partial f, partial x". Shorthand: f_x and f_y. The curly d is a warning to the reader that other variables exist and are being held fixed.

In practice we never take these limits by hand. Treat every other variable as a **constant**, then use the rules from earlier in these notes.

Worked example: the bowl. For $f(x, y) = x^2 + y^2$:

$$\frac{\partial f}{\partial x} = 2x + 0 = 2x \quad (y^2 \text{ is a constant}), \qquad \frac{\partial f}{\partial y} = 0 + 2y = 2y \quad (x^2 \text{ is a constant}).$$

Worked example: mixed terms. For $f(x, y) = 3x^2y + 5y^3 - 2x$:

- With respect to x, y is a constant: in $3x^2y$ the $3y$ rides along as a constant multiple, $5y^3$ is a constant added, and $-2x$ is a line. So $f_x = 6xy + 0 - 2 = 6xy - 2$.
- With respect to y, x is a constant: in $3x^2y$ the $3x^2$ rides along, $5y^3 \to 15y^2$, and $-2x$ is a constant added. So $f_y = 3x^2 + 15y^2$.

Worked example: drug concentration. For $C(D, V) = \dfrac{D}{V} = D\,V^{-1}$:

$$\frac{\partial C}{\partial D} = \frac{1}{V}, \qquad \frac{\partial C}{\partial V} = D \cdot (-1)V^{-2} = -\frac{D}{V^2}.$$

For a dose of $D = 500$ mg spread through $V = 40$ L (so $C = 12.5$ mg/L): $\frac{\partial C}{\partial D} = \frac{1}{40} = 0.025$ mg/L per mg, so each extra mg of dose adds 0.025 mg/L; and $\frac{\partial C}{\partial V} = -\frac{500}{1600} \approx -0.31$ mg/L per L, so a patient with one more liter of volume ends up about 0.31 mg/L lower.

Worked example: the chain rule still applies. For $f(x, y) = e^{xy}$, with y frozen the exponent xy is a constant times x, with derivative y:

$$\frac{\partial f}{\partial x} = e^{xy} \cdot y = y\,e^{xy}, \qquad \frac{\partial f}{\partial y} = e^{xy} \cdot x = x\,e^{xy}.$$

Watch out: "constant" does not mean "zero"

When finding $\frac{\partial}{\partial x}$ of x^2y, the y is held fixed, not deleted: the answer is $2xy$, not $2x$ and not 0. A frozen variable behaves like the number 5 would. Only a term with **no** x at all, like $5y^3$ above, differentiates to 0.

7.4 What a Partial Means: BMI

Body mass index is

$$B(w, h) = \frac{w}{h^2} = w\,h^{-2},$$

with weight w in kg and height h in m.

- Freeze h: B is w times the constant h^{-2}, so $\dfrac{\partial B}{\partial w} = h^{-2} = \dfrac{1}{h^2}$.
- Freeze w: w rides along, and the power rule on h^{-2} gives $\dfrac{\partial B}{\partial h} = w \cdot (-2)h^{-3} = -\dfrac{2w}{h^3}$.

For a patient with $w = 70$ kg and $h = 1.75$ m (so $B \approx 22.9$):

- $\dfrac{\partial B}{\partial w} = \dfrac{1}{1.75^2} \approx 0.33$ BMI points per kg: one more kg raises this patient's BMI by about a **third** of a point.
- $\dfrac{\partial B}{\partial h} = -\dfrac{2 \cdot 70}{1.75^3} \approx -26$ BMI points per m. A meter is a huge change in height, so divide by 100: about -0.26 per cm. One more cm of height, at the same weight, lowers BMI by about a **quarter** of a point.

Units. A partial derivative is in output units **per unit of the input that moved**: BMI points per kg for the first, per m (or per cm) for the second.

The partials also depend on *where* you are. A shorter patient, $h = 1.5$ m, has $\frac{\partial B}{\partial w} = \frac{1}{2.25} \approx 0.44$: each kg moves a shorter person's BMI more.

Exercise 7.1

1. Find both partial derivatives of $f(x, y) = x^3 + 4xy - y^2$.
2. Find both partial derivatives of $g(x, y) = x\, e^y$.
3. For a patient with $w = 80$ kg and $h = 1.6$ m, find $\frac{\partial B}{\partial w}$ and say in words what it means.

Solution to Exercise 7.1

1. With y frozen: $3x^2 + 4y - 0$, so $f_x = 3x^2 + 4y$. With x frozen: $0 + 4x - 2y$, so $f_y = 4x - 2y$. In $4xy$, the frozen variable rides along as a constant multiple in each case.
2. With y frozen, e^y is a constant multiple of x: $g_x = e^y$. With x frozen, x is a constant multiple of e^y, which is its own derivative: $g_y = x\, e^y$.
3. $\frac{\partial B}{\partial w} = \frac{1}{h^2} = \frac{1}{1.6^2} = \frac{1}{2.56} \approx 0.39$ BMI points per kg. At this height, each additional kg of weight raises BMI by about 0.39 points.

8 Review

What we covered: from secant to tangent

1. The **average rate of change** from x_1 to x_2 is $\frac{f(x_2) - f(x_1)}{x_2 - x_1}$: the slope of a **secant line**. Its units are output per input.
2. The **difference quotient** $\frac{f(x+h) - f(x)}{h}$ is the same slope, from x to $x + h$.
3. Setting $h = 0$ gives $0/0$. Instead take a **limit**: let h get as close to 0 as we like, without reaching it. Simplify first, then let h shrink.
4. The **derivative** $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$ is the slope of the **tangent line**: the **instantaneous** rate of change. Also written $\frac{df}{dx}$, $\frac{d}{dx} f(x)$, and, for time, $\dot{x}$.
5. The sign of f' says rising (> 0), falling (< 0), or flat ($= 0$).

What we covered: the rules

6. From the definition: $c \to 0$, $x \to 1$, $mx+b \to m$, $x^2 \to 2x$, $\sin x \to \cos x$, $\frac{1}{x} \to -\frac{1}{x^2}$.
7. **Power rule**: $x^n \to n\,x^{n-1}$. Rewrite roots and fractions as powers first.
8. Constants **ride along**; sums and differences **split**.
9. **Product rule**: $(fg)' = f'g + fg'$. **Quotient rule**: $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$.
10. The table: $e^x \to e^x$, $\ln x \to \frac{1}{x}$, $a^x \to a^x \ln a$, $\log_a x \to \frac{1}{x \ln a}$, $\sin x \to \cos x$, $\cos x \to -\sin x$.
11. **Chain rule**: for a composition $h(x) = f(g(x))$, $h'(x) = f'(g(x)) \cdot g'(x)$: the outer's derivative at the inner, times the inner's derivative.

What we covered: two variables

12. A function $f(x, y)$ of two variables has a **surface** for a graph (Unit 1-3).
13. A **partial derivative** freezes the other variable and differentiates with the same rules: $\frac{\partial f}{\partial x}$ is the slope of the slice in the x direction, in output units per input unit.

Notebooks:

- `U3-2_Limits-1_ApproachingALimit.ipynb`
- `U3-3_Derivatives-1_SecantToTangent.ipynb`
- `U3-3_Derivatives-2_Rules.ipynb`
- `U3-6_Multivariable-1_Surface3D.ipynb`
- `U3-6_Multivariable-2_PartialDerivatives.ipynb`

Previous: Unit 2-5, **Determinants and Inverses**.

Next: Unit 3-2, **Optimization and Linear Regression**: using derivatives to find the lowest point of a model's error, and fitting a line to data two ways.